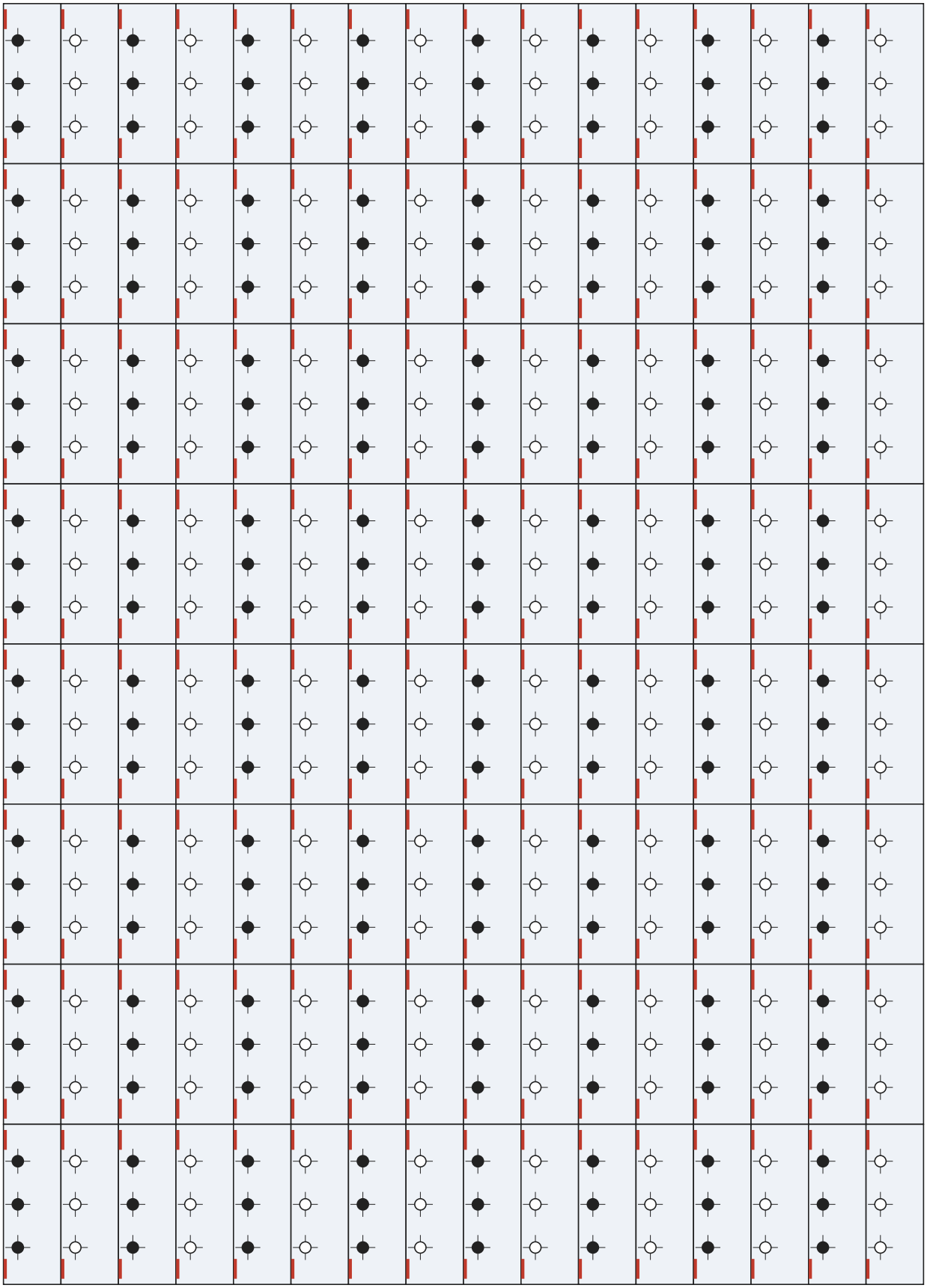


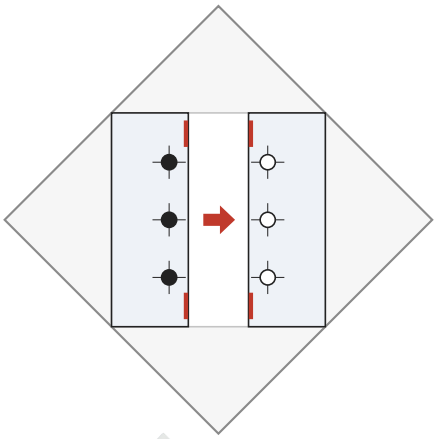
SkinKit magnet sheet: how to use it

For US Letter paper (8.5 x 11 in). The other side is the pattern.

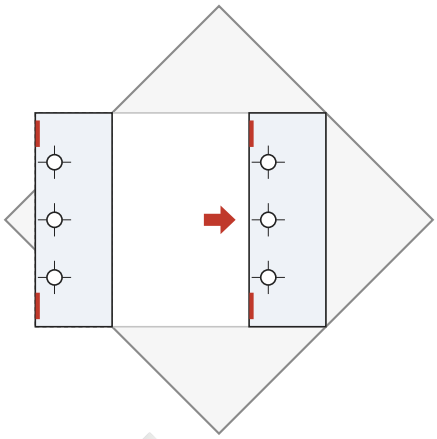
LETTER SIZE (8.5 x 11 in)
PRINT AT 100% / ACTUAL SIZE
Not "Fit to page". Check the rulers below.

1. Make a second Letter-size sheet of skin cloth, the same way as the first. Keep it whole and cover it with double-sided tape.
2. Lay it over the pattern side of this sheet. Skin cloth is see-through, so the pattern shows. Tape all four edges down with masking or washi tape so nothing slides. The pattern leaves a 2 cm border, because the edges of a sheet often get damaged.
3. Put a magnet on every dot with tweezers. Black dots: all one face up. White dots: all flipped, so the pole is opposite. Which face is N does not matter, but keep it the same on every piece.
4. Cut the sheet into strips along the solid lines, through the skin cloth and the paper together. The paper keeps the lines visible and the skin cloth flat. Then peel the paper off the back.
5. Stick one strip at each end of a diamond (Cross: one per arm). A flat end takes black at the input and white at the output. A flap folds over when you use it, which flips its magnets, so a flap takes the opposite colour: white on an input flap, black on an output flap. Line the plain edge (no red marks) up with the edge of the inner square: on a flap the strip covers the flap, on a flat end it sits just inside the square. See below, real size. Red arrows point the way the signal flows.

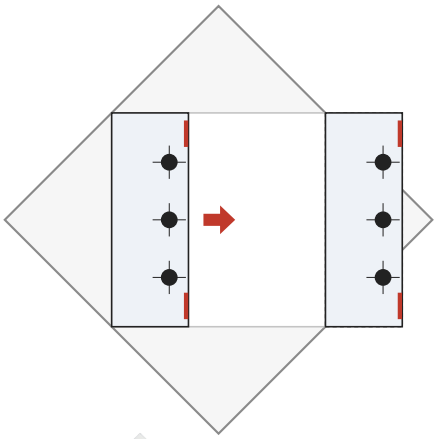
Where each piece goes (real size)



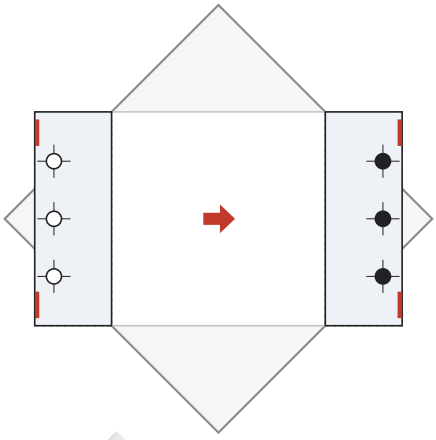
Flat
Black strip just inside the square at the input end, white strip just inside at the output end.



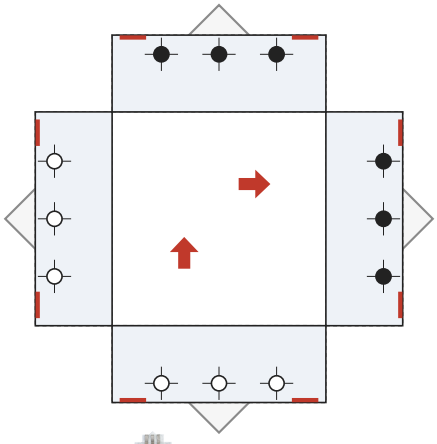
Flap-in
White on the input flap (it flips when the flap folds), white just inside the square at the output end.



Flap-out
Black just inside the square at the input end, black on the output flap (it flips when the flap folds).



Double flaps
White on the input flap, black on the output flap. Both flip when the flaps fold.



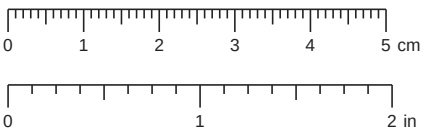
Cross
One strip per arm: white on the input arms, black on the output arms. The arms fold, which flips them.

Key

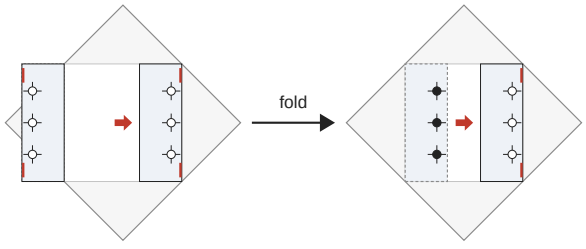
- magnet, one face up
- magnet, flipped
- flap outline (folds here)
- dots sit 2.54 mm from this edge
- signal direction

Strip 10.16 x 28.28 mm, the size of a flap. Inner square 28.28 mm (diamond with 40 mm sides). Magnets 2 mm, 7.62 mm (0.3 in) apart, like on a circuit module.

Scale check



Why a flap takes the opposite colour (Flap-in shown, 55% size)



While you make it: flap open — **When you use it: flap folded under**

Side view (not to scale)



Open: the strip sits on top of the flap with the white side of each magnet up.



Folded: the flap turns over, so the same magnet is now black side up, where a flat end's black magnets sit.

Strips per module: Flat 1 black + 1 white, Flap-in 2 white, Flap-out 2 black, Double flaps 1 + 1, Cross 2 + 2. 64 of each colour per sheet: make Flap-in and Flap-out in equal numbers.